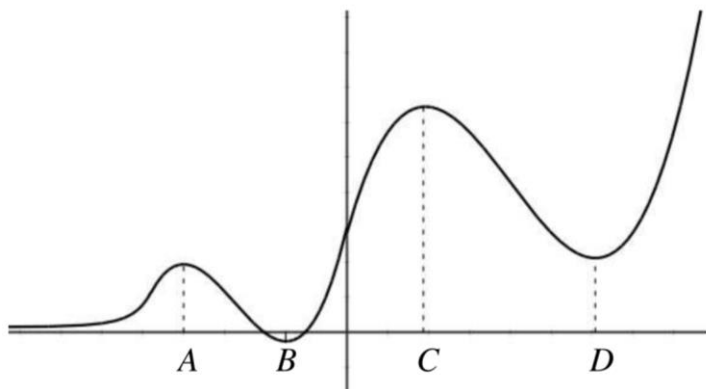


1. The derivative of a function $f(x)$ defined on $(-\infty, \infty)$ is $f'(x) = x^3 - 9x$, then which of the following statements is true?
- (a) The function $f(x)$ has a local maximum at $x = 0$.
(b) The function $f(x)$ has an absolute minimum at $x = 0$.
(c) The function $f(x)$ has a local maximum at $x = 3$ and $x = -3$.
(d) The function $f(x)$ has an absolute maximum at $x = 3$ and $x = -3$.
(e) None of the above.

2. Find all intervals where the function $f(x) = \frac{1}{4}x^4 - x^3 + x^2$ increasing?

3. The graph of $f'(x)$ is shown. On which of the following intervals is f concave downward? (You are being asked about f , but you are shown the graph of the **derivative** of f .)



4. Find the value of a so that $f(x) = ax^3 + 2x^2 + 2x + 1$ has an inflection point when $x = 1$.

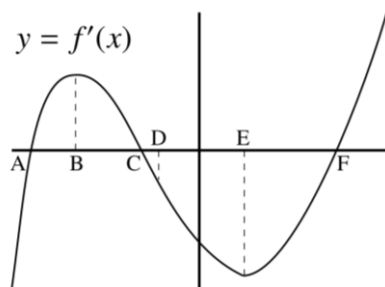
5. The derivative of a function $g(x)$ is given by $g'(x) = -7(x + 3)^2(x - 1)(x - 5)$. Find the x -coordinates for each local maximum and local minimum of $g(x)$, if any.

6. If $f(x)$ has a local maximum at $x = a$, then whether each of the following is > 0 , < 0 or $= 0$:

i. $f'(a)$

ii. $f''(a)$

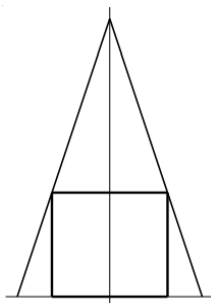
7. The graph of the derivative of a function $f(x)$ is given. (So the graph of $y = f'(x)$ is shown).



The function $f(x)$ is increasing on the interval(s)

The function $f(x)$ is concave up on the interval(s)

8. Find the maximum area of a rectangle that is inside of the triangle formed by the x -axis and the lines $y = -3x + 12$ and $y = 3x + 12$. The base of the rectangle is on the x -axis and the two upper vertices are on the lines $y = -3x + 12$ and $y = 3x + 12$ as shown in the illustration.



9. Find the value(s) of c that is guaranteed by the Mean Value Theorem for the function $f(x) = x^3 - \frac{1}{2}x^2$ on the interval $[-2, 2]$.

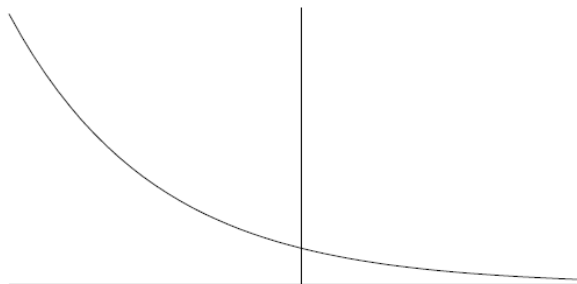
10. Find the x-coordinate of each inflection point of the function $f(x) = x^4 - 54x^2 - 8x + 10$.

11. The graph for the function $f(x)$ is shown below. State whether each of the following is > 0 (greater than 0), < 0 (less than 0) or $= 0$ (equal to 0) for all values of x .

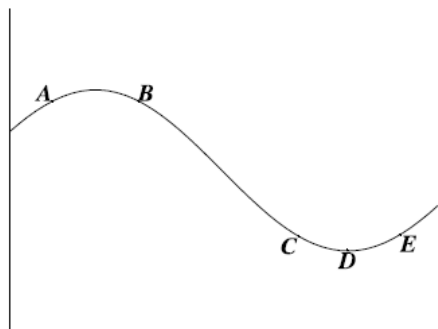
i. $f(x)$

ii. $f'(x)$

iii. $f''(x)$



12. Given the graph of $f(x)$, on which point(s) on the graph are $f'(x)$ and $f''(x)$ both negative?



13. The derivative of the function is given by $f'(x) = \frac{1}{1+x^2} - e^x + \frac{5}{x} + 1$. Find the general antiderivative $f(x)$.

(a) $f(x) = \tan^{-1} x - e^x + \frac{5}{x^2} + x + C$

(b) $f(x) = \tan^{-1} x - e^x + 5 \ln|x| + x + C$

(c) $f(x) = \frac{x}{1+x^2} - e^x + \frac{5}{x^2} + x + C$

(d) $f(x) = \frac{1}{1+x^3} - e^x + \frac{5}{x^2} + x + C$

(e) $f(x) = \frac{1}{1+\frac{x^3}{3}} - e^x + \frac{5}{x^2} + x + C$

14. If $f'(x) = e^x + 12x^2 - 4$ and $f(0) = -2$ then $f(x)$ equals

- (a) $e^x + 12x^3 - 4x - 3$
- (b) $e^x + 4x^3 - 4x - 2$
- (c) $e^x + 24x - 2$
- (d) $e^x + 4x^3 - 4x - 3$
- (e) $e^x + 12x^3 - 4x$

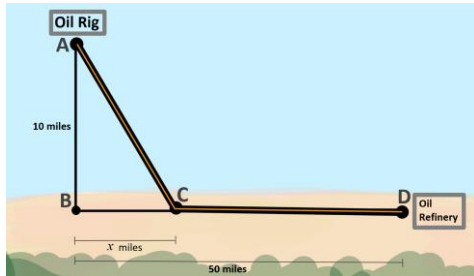
15. $\int 2x^2 + \sqrt[3]{x} - 1 \, dx =$

- (a) $\frac{2}{3}x^3 + \frac{3}{4}x^{\frac{4}{3}} - x + C$
- (b) $\frac{3}{2}x^3 + \frac{4}{3}x^{\frac{4}{3}} - x + C$
- (c) $\frac{x^3}{3} + \frac{x^{\frac{4}{3}}}{4} - x + C$
- (d) $x^3 + x^{\frac{4}{3}} - x + C$
- (e) $\frac{2}{3}x^3 + \frac{\sqrt[4]{x}}{4} - x + C$

16. Find the Absolute Maximum and Absolute Minimum values of the function $f(x) = 2x^3 - 3x^2 - 36x$ on the interval $[-3, 4]$.

17. For the function $f(x) = x^4 - 4x^3 - 2x + 8$, Find the intervals on the x-axis where the function is concave upward and where it is concave downward. Use interval notation (a,b) for your answers.

18. An offshore oil rig is located in the ocean at a point A, which is 10 mi from the closest point B on the shore, on a straight shoreline. The oil is to be piped to a point D on the shore that is 50 miles away from point B. The cost of laying pipe is \$2 million per mile under water and \$1.5 million per mile over land. The most cost-effective way of constructing the pipeline would be to connect the oil rig to a point C on the shore between B and D (i.e. segment AC is completely underwater) and then build the remaining segment between over land between C and D. If point C is x miles away from point B, find the length of x that will minimize construction cost for the entire pipeline.



The cost of the pipeline can be expressed as the function

$$C(x) = (\text{length of AC}) \cdot (\text{cost of underwater pipe/mile}) + (\text{length of CD}) \cdot (\text{cost of overland pipe/mile})$$

- i. Give an expression for AC in terms of x

- ii. Give an expression for CD in terms of x

- iii. Find the length of x (in miles) that will minimize the cost.

(Round this value to 3 decimal places)

19. Starting with $x_0 = 6$ find the first, second and third approximation to the root of the equation $x^2 - 2x = 15$. Do not round any values until the end, round your final answer(s) to 8 decimal places.

n	x_n	$f(x_n)$	$f'(x_n)$	$x_{n+1} = x_n - \frac{f(x_n)}{f'(x_n)}$
0				
1				
2				
3				

20. A stone is dropped from the edge of a roof, and hits the ground with a velocity of -49 meters per second. Assume the acceleration due to gravity is -9.8 meters per second². How high (in meters) is the roof?